Risultati della campagna intermedia: stati di Fock, Cat e Squeezed

Tabella 1 — Simulazioni principali con filtro passa-basso causale ($\omega_{\mathbf{cut}} = 3.0\,\omega_c$, blocchi `main_000`, `main_002`–`main_004`, `main_006`–`main_008`, `main_011`–`main_014`).

Esperimento	Sistema	Successi (valid. num.)	P_{target} med. (continua)	P_{target} [min, max]	Fluenza $\mathcal{F}_{\text{mod}}$ tot (max ch)	Costo paper C_2/ω_c^2 tot (max ch)	$\text{SR}_{\max}$ / $\omega_{99\%}$ (mediana)	Iter. GRAPE med. [min, max]
Fock $\lvert n=6\rangle$, $T = 10\,\tau_s$ filtro 3.0, dim = 60 (`main_000`)	1q	0/8 (0/8)	0.717317	[0.617159, 0.748426]	34.06 (34.06)	17.03 (17.03)	3.78 / 4.74	298 [206, 300]
	2q_common	5/8 (7/8)	**0.993292**	[0.973599, 0.999936]	53.03 (26.51)	26.51 (13.26)	3.77 / 5.46	300 [300, 300]
	2q_independent	**8/8** (8/8)	**0.999942**	[0.999874, 0.999948]	48.16 (25.29)	24.08 (12.65)	3.73 / 4.98	300 [197, 300]
	2q_norm_common	0/8 (0/8)	0.651781	[0.525758, 0.917784]	79.07 (39.54)	39.54 (19.77)	3.78 / 4.62	300 [258, 300]
	2q_norm_independent	0/8 (0/8)	0.738014	[0.384006, 0.768554]	73.13 (37.44)	36.56 (18.72)	3.78 / 4.74	300 [300, 300]
Fock $\lvert n=10\rangle$, $T = 10\,\tau_s$ filtro 3.0, dim = 60 (`main_002`)	1q	0/8 (0/8)	0.302737	[0.158032, 0.400833]	39.11 (39.11)	19.56 (19.56)	3.78 / 4.74	208 [151, 300]
	2q_common	0/8 (0/8)	0.882637	[0.763084, 0.947159]	59.26 (29.63)	29.63 (14.81)	3.78 / 5.04	300 [167, 300]
	2q_independent	0/8 (0/8)	0.921985	[0.877577, 0.975032]	65.43 (34.17)	32.72 (17.08)	3.78 / 5.22	300 [300, 300]
	2q_norm_common	0/8 (0/8)	0.270133	[0.107894, 0.490162]	75.16 (37.58)	37.58 (18.79)	3.78 / 4.44	256 [187, 300]
	2q_norm_independent	0/8 (0/8)	0.354498	[0.233794, 0.466442]	81.09 (43.26)	40.55 (21.63)	3.78 / 4.68	300 [300, 300]
Fock $\lvert n=10\rangle$, $T = 15\,\tau_s$ filtro 3.0, dim = 60 (`main_003`)	1q	0/8 (0/8)	0.937283	[0.872441, 0.973582]	52.27 (52.27)	26.13 (26.13)	3.78 / 4.84	300 [219, 300]
	2q_common	**8/8** (8/8)	**0.999946**	[0.999808, 0.999949]	62.98 (31.49)	31.49 (15.75)	3.69 / 5.04	264 [196, 300]
	2q_independent	**8/8** (8/8)	**0.999945**	[0.999943, 0.999946]	65.08 (33.66)	32.54 (16.83)	3.68 / 4.72	204 [169, 300]
	2q_norm_common	0/8 (0/8)	0.838046	[0.574137, 0.947635]	95.02 (47.51)	47.51 (23.76)	3.78 / 4.84	300 [300, 300]
	2q_norm_independent	1/8 (1/8)	0.979306	[0.810678, 0.992984]	110.74 (56.80)	55.37 (28.40)	3.78 / 4.80	300 [300, 300]
Fock $\lvert n=10\rangle$, $T = 20\,\tau_s$ filtro 3.0, dim = 60 (`main_004`)	1q	**8/8** (8/8)	**0.999948**	[0.999945, 0.999950]	52.01 (52.01)	26.01 (26.01)	3.76 / 4.68	182 [152, 225]
	2q_common	**8/8** (8/8)	**0.999946**	[0.999946, 0.999948]	65.51 (32.75)	32.75 (16.38)	3.56 / 4.32	186 [127, 230]
	2q_independent	**8/8** (8/8)	**0.999946**	[0.999944, 0.999948]	54.25 (30.23)	27.13 (15.12)	3.61 / 4.29	176 [118, 218]
	2q_norm_common	6/8 (6/8)	**0.998454**	[0.926864, 0.999860]	117.09 (58.55)	58.55 (29.27)	3.78 / 4.98	300 [300, 300]
	2q_norm_independent	**8/8** (8/8)	**0.999945**	[0.999937, 0.999949]	103.18 (54.42)	51.59 (27.21)	3.76 / 4.77	282 [198, 300]
Cat $\alpha = 2.0$, $T = 10\,\tau_s$ filtro 3.0, dim = 48 (`main_006`)	1q	1/8 (6/8)	0.977199	[0.951746, 0.992273]	32.45 (32.45)	16.22 (16.22)	3.78 / 4.80	300 [300, 300]
	2q_common	**8/8** (8/8)	**0.999431**	[0.997890, 0.999856]	45.67 (22.84)	22.84 (11.42)	3.76 / 5.28	300 [300, 300]
	2q_independent	**8/8** (8/8)	**0.999838**	[0.999489, 0.999906]	47.28 (25.06)	23.64 (12.53)	3.77 / 4.86	300 [300, 300]
	2q_norm_common	0/8 (0/8)	0.883836	[0.873494, 0.919944]	64.95 (32.47)	32.47 (16.24)	3.78 / 4.86	300 [300, 300]
	2q_norm_independent	2/8 (3/8)	0.979713	[0.956176, 0.998721]	65.29 (33.94)	32.64 (16.97)	3.78 / 5.04	300 [300, 300]
Cat $\alpha = 2.0$, $T = 15\,\tau_s$ filtro 3.0, dim = 48 (`main_007`)	1q	**8/8** (8/8)	**0.999949**	[0.999947, 0.999951]	32.20 (32.20)	16.10 (16.10)	3.68 / 4.68	223 [127, 273]
	2q_common	**8/8** (8/8)	**0.999949**	[0.999948, 0.999950]	46.99 (23.50)	23.50 (11.75)	3.07 / 4.08	238 [187, 299]
	2q_independent	**8/8** (8/8)	**0.999949**	[0.999862, 0.999950]	38.61 (23.16)	19.30 (11.58)	3.20 / 4.16	294 [233, 300]
	2q_norm_common	**8/8** (8/8)	**0.999399**	[0.995952, 0.999910]	83.96 (41.98)	41.98 (20.99)	3.74 / 4.68	300 [300, 300]
	2q_norm_independent	**8/8** (8/8)	**0.999883**	[0.999680, 0.999947]	79.29 (42.38)	39.64 (21.19)	3.73 / 4.84	300 [300, 300]
Cat $\alpha = 2.0$, $T = 20\,\tau_s$ filtro 3.0, dim = 48 (`main_008`)	1q	**8/8** (8/8)	**0.999948**	[0.999947, 0.999949]	30.65 (30.65)	15.33 (15.33)	3.52 / 4.14	170 [122, 235]
	2q_common	**8/8** (8/8)	**0.999949**	[0.999947, 0.999950]	43.38 (21.69)	21.69 (10.84)	2.67 / 3.75	215 [185, 300]
	2q_independent	**8/8** (8/8)	**0.999949**	[0.999948, 0.999950]	35.31 (18.83)	17.66 (9.42)	2.65 / 3.84	241 [225, 290]
	2q_norm_common	**8/8** (8/8)	**0.999947**	[0.999920, 0.999949]	72.35 (36.18)	36.18 (18.09)	3.65 / 4.20	278 [198, 300]
	2q_norm_independent	**8/8** (8/8)	**0.999948**	[0.999840, 0.999950]	64.54 (35.00)	32.27 (17.50)	3.69 / 4.02	292 [174, 300]
Squeezed 3 dB, $T = 10\,\tau_s$ filtro 3.0, dim = 48 (`main_011`)	1q	**8/8** (8/8)	**0.999953**	[0.999950, 0.999964]	5.54 (5.54)	2.77 (2.77)	1.67 / 3.60	34 [29, 39]
	2q_common	**8/8** (8/8)	**0.999951**	[0.999950, 0.999953]	13.36 (6.68)	6.68 (3.34)	1.45 / 3.60	35 [26, 39]
	2q_independent	**8/8** (8/8)	**0.999951**	[0.999950, 0.999953]	8.40 (4.74)	4.20 (2.37)	1.53 / 3.54	43 [39, 57]
	2q_norm_common	**8/8** (8/8)	**0.999951**	[0.999950, 0.999952]	12.54 (6.27)	6.27 (3.14)	1.76 / 3.60	37 [31, 49]
	2q_norm_independent	**8/8** (8/8)	**0.999953**	[0.999950, 0.999955]	9.39 (5.48)	4.70 (2.74)	1.40 / 3.60	46 [40, 52]
Squeezed 3 dB, $T = 20\,\tau_s$ filtro 3.0, dim = 48 (`main_012`)	1q	**8/8** (8/8)	**0.999953**	[0.999950, 0.999965]	6.77 (6.77)	3.38 (3.38)	1.39 / 3.66	28 [23, 39]
	2q_common	**8/8** (8/8)	**0.999953**	[0.999950, 0.999961]	7.83 (3.91)	3.91 (1.96)	1.22 / 3.09	26 [15, 34]
	2q_independent	**8/8** (8/8)	**0.999952**	[0.999950, 0.999955]	10.23 (5.60)	5.12 (2.80)	1.39 / 3.12	33 [25, 39]
	2q_norm_common	**8/8** (8/8)	**0.999951**	[0.999950, 0.999952]	12.88 (6.44)	6.44 (3.22)	1.28 / 3.36	42 [31, 56]
	2q_norm_independent	**8/8** (8/8)	**0.999951**	[0.999950, 0.999953]	12.24 (6.41)	6.12 (3.20)	1.52 / 3.24	47 [45, 56]
Squeezed 8 dB, $T = 10\,\tau_s$ filtro 3.0, dim = 48 (`main_013`)	1q	**8/8** (8/8)	**0.998505**	[0.998025, 0.999255]	27.64 (27.64)	13.82 (13.82)	3.75 / 5.10	300 [295, 300]
	2q_common	**8/8** (8/8)	**0.999404**	[0.999055, 0.999621]	36.25 (18.12)	18.12 (9.06)	3.71 / 5.16	300 [300, 300]
	2q_independent	**8/8** (8/8)	**0.999285**	[0.998880, 0.999644]	33.79 (17.98)	16.90 (8.99)	3.68 / 5.28	300 [300, 300]
	2q_norm_common	**8/8** (8/8)	**0.998954**	[0.998128, 0.999326]	50.83 (25.42)	25.42 (12.71)	3.69 / 4.92	300 [300, 300]
	2q_norm_independent	**8/8** (8/8)	**0.999400**	[0.998095, 0.999610]	50.82 (26.42)	25.41 (13.21)	3.70 / 5.10	300 [300, 300]
Squeezed 8 dB, $T = 20\,\tau_s$ filtro 3.0, dim = 48 (`main_014`)	1q	**8/8** (8/8)	**0.999627**	[0.999159, 0.999809]	27.72 (27.72)	13.86 (13.86)	3.67 / 4.68	300 [300, 300]
	2q_common	**8/8** (8/8)	**0.999751**	[0.999635, 0.999842]	32.20 (16.10)	16.10 (8.05)	3.19 / 4.92	300 [300, 300]
	2q_independent	**8/8** (8/8)	**0.999607**	[0.999461, 0.999771]	30.22 (15.40)	15.11 (7.70)	3.41 / 4.92	300 [300, 300]
	2q_norm_common	**8/8** (8/8)	**0.999647**	[0.999511, 0.999841]	55.25 (27.63)	27.63 (13.81)	3.55 / 3.96	300 [300, 300]
	2q_norm_independent	**8/8** (8/8)	**0.999636**	[0.999482, 0.999692]	54.01 (30.02)	27.00 (15.01)	3.50 / 4.02	300 [300, 300]

Tabella 2 — Simulazioni di controllo senza filtro (ω_{cut} = null, blocchi `main_001`, `main_005`, `main_009`, `main_010`, `main_015`, `main_016`).

Esperimento	Sistema	Successi (valid. num.)	P_{target} med. (continua)	P_{target} [min, max]	Fluenza $\mathcal{F}_{\mathrm{mod}}$ tot (max ch)	Costo paper C_2/ω_c^2 tot (max ch)	$\mathrm{SR}_{\max}$ / $\omega_{99\%}$ (mediana)	Iter. GRAPE med. [min, max]
Fock $\lvert n=6\rangle$, $T=10\,\tau_s$ senza filtro, dim = 60 (`main_001`)	`1q`	1/8 (1/8)	0.981119	[0.956183, 0.999322]	49.20 (49.20)	24.60 (24.60)	4.77 / 8.16	300 [235, 300]
	`2q_common`	**8/8** (8/8)	**0.999937**	[0.999917, 0.999939]	53.70 (26.85)	26.85 (13.42)	4.76 / 7.80	218 [179, 300]
	`2q_independent`	**8/8** (8/8)	**0.999938**	[0.999929, 0.999941]	54.19 (30.55)	27.09 (15.27)	4.76 / 8.16	176 [130, 214]
	`2q_norm_common`	0/8 (0/8)	0.828066	[0.777615, 0.936965]	97.49 (48.75)	48.75 (24.37)	4.77 / 7.56	300 [300, 300]
	`2q_norm_independent`	0/8 (0/8)	0.976268	[0.887827, 0.995310]	103.93 (54.05)	51.96 (27.03)	4.77 / 7.44	300 [300, 300]
Fock $\lvert n=10\rangle$, $T=10\,\tau_s$ senza filtro, dim = 60 (`main_005`)	`1q`	0/8 (0/8)	0.687505	[0.603246, 0.724137]	52.88 (52.88)	26.44 (26.44)	4.77 / 7.80	190 [157, 300]
	`2q_common`	3/8 (3/8)	**0.994516**	[0.959896, 0.999734]	79.74 (39.87)	39.87 (19.93)	4.77 / 7.86	300 [300, 300]
	`2q_independent`	**8/8** (8/8)	**0.999737**	[0.998240, 0.999903]	78.60 (40.65)	39.30 (20.33)	4.77 / 8.04	300 [300, 300]
	`2q_norm_common`	0/8 (0/8)	0.502739	[0.383340, 0.638977]	100.73 (50.37)	50.37 (25.18)	4.77 / 7.68	298 [240, 300]
	`2q_norm_independent`	0/8 (0/8)	0.653033	[0.565350, 0.806457]	104.43 (53.58)	52.21 (26.79)	4.77 / 7.92	300 [300, 300]
Cat $\alpha=2.0$, $T=10\,\tau_s$ senza filtro, dim = 48 (`main_009`)	`1q`	**8/8** (8/8)	**0.999789**	[0.997088, 0.999928]	43.56 (43.56)	21.78 (21.78)	4.77 / 7.74	300 [300, 300]
	`2q_common`	7/8 (7/8)	**0.999931**	[0.999776, 0.999946]	54.80 (27.40)	27.40 (13.70)	4.65 / 7.14	294 [209, 300]
	`2q_independent`	**8/8** (8/8)	**0.999941**	[0.999934, 0.999944]	45.76 (25.09)	22.88 (12.55)	4.70 / 7.80	285 [191, 300]
	`2q_norm_common`	3/8 (3/8)	0.989409	[0.922015, 0.997865]	89.97 (44.98)	44.98 (22.49)	4.77 / 7.74	300 [300, 300]
	`2q_norm_independent`	**8/8** (8/8)	**0.999779**	[0.999151, 0.999888]	79.36 (41.00)	39.68 (20.50)	4.77 / 7.98	300 [300, 300]
Cat $\alpha=2.0$, $T=15\,\tau_s$ senza filtro, dim = 48 (`main_010`)	`1q`	**8/8** (8/8)	**0.999942**	[0.999941, 0.999943]	32.73 (32.73)	16.37 (16.37)	4.56 / 7.76	130 [125, 175]
	`2q_common`	**8/8** (8/8)	**0.999944**	[0.999940, 0.999945]	42.10 (21.05)	21.05 (10.52)	3.55 / 6.72	189 [140, 253]
	`2q_independent`	**8/8** (8/8)	**0.999944**	[0.999940, 0.999946]	35.87 (19.65)	17.94 (9.83)	3.48 / 6.60	220 [131, 270]
	`2q_norm_common`	7/8 (7/8)	**0.999918**	[0.999815, 0.999944]	88.57 (44.28)	44.28 (22.14)	4.72 / 7.52	300 [201, 300]
	`2q_norm_independent`	**8/8** (8/8)	**0.999942**	[0.999888, 0.999944]	73.66 (44.05)	36.83 (22.03)	4.63 / 6.64	250 [171, 300]
Squeezed 3 **dB**, $T=10\,\tau_s$ senza filtro, dim = 48 (`main_015`)	`1q`	**8/8** (8/8)	**0.999954**	[0.999950, 0.999959]	5.84 (5.84)	2.92 (2.92)	1.72 / 3.96	31 [24, 34]
	`2q_common`	**8/8** (8/8)	**0.999951**	[0.999950, 0.999959]	12.78 (6.39)	6.39 (3.20)	1.76 / 4.14	35 [30, 44]
	`2q_independent`	**8/8** (8/8)	**0.999951**	[0.999950, 0.999956]	8.57 (4.87)	4.28 (2.44)	1.52 / 3.96	38 [30, 46]
	`2q_norm_common`	**8/8** (8/8)	**0.999952**	[0.999950, 0.999955]	11.60 (5.80)	5.80 (2.90)	1.86 / 3.78	37 [32, 47]
	`2q_norm_independent`	**8/8** (8/8)	**0.999950**	[0.999950, 0.999955]	9.98 (5.79)	4.99 (2.90)	1.72 / 3.84	40 [36, 48]
Squeezed 8 **dB**, $T=10\,\tau_s$ senza filtro, dim = 48 (`main_016`)	`1q`	**8/8** (8/8)	**0.999550**	[0.999173, 0.999677]	33.96 (33.96)	16.98 (16.98)	4.77 / 8.10	300 [300, 300]
	`2q_common`	6/8 (6/8)	**0.999720**	[0.999618, 0.999906]	40.60 (20.30)	20.30 (10.15)	4.72 / 7.80	300 [300, 300]
	`2q_independent`	**8/8** (8/8)	**0.999752**	[0.999635, 0.999897]	36.58 (20.85)	18.29 (10.42)	4.77 / 8.10	300 [300, 300]
	`2q_norm_common`	**8/8** (8/8)	**0.999610**	[0.999480, 0.999791]	64.46 (32.23)	32.23 (16.11)	4.77 / 7.68	300 [300, 300]
	`2q_norm_independent`	**8/8** (8/8)	**0.999727**	[0.999616, 0.999795]	60.13 (31.80)	30.07 (15.90)	4.77 / 8.04	300 [300, 300]

- **Sistemi confrontati:** `1q` (1 qubit con $g = 0.3\,\omega_c$); `2q_common` (2 qubit con $g_1 = g_2 = 0.3\,\omega_c$, 1 canale comune); `2q_independent` (2 qubit con $g_1 = g_2 = 0.3\,\omega_c$, 2 canali indipendenti); `2q_norm_common` (2 qubit con $g_1 = g_2 = 0.3/\sqrt{2}\,\omega_c$ tali che $g_{\mathrm{eff}} = \sqrt{g_1^2 + g_2^2} = 0.3\,\omega_c$, 1 canale comune); `2q_norm_independent` (2 qubit con $g_1 = g_2 = 0.3/\sqrt{2}\,\omega_c$, 2 canali indipendenti).
- **Fluenza e costo energetico:** $\mathcal{F}_{\mathrm{mod,tot}} = \sum_j \int_0^T \Omega_{D,j}(t)^2\,dt$ e costo $C_2/\omega_c^2 = \frac{1}{2}\mathcal{F}_{\mathrm{mod}}$.
- **Slew rate e banda:** $\mathrm{SR}_{\max} = \max_{j,t} |\dot{\Omega}_{D,j}(t)|/\omega_c^2$ (massima rapidità di variazione del drive) e $\omega_{99\%}/\omega_c$ (frequenza entro cui è contenuto il 99% della potenza spettrale AC del controllo applicato).